MediConnect

Product Requirements Document — Telehealth Patient Portal (v2.4)

Status: Approved for Development | Owner: Clinical Platform Team | Version 2.4 | Classification: Internal / HIPAA-Sensitive

1. Executive Summary

MediConnect is a telehealth platform that connects patients with licensed clinicians for virtual consultations, prescription management, and ongoing care coordination. Patients book appointments, complete intake forms, join encrypted video visits, and message their care team, while clinicians document visits, e-prescribe medication, and bill insurance directly from the platform. This document defines requirements for the v2.4 release, which adds direct EHR interoperability with major hospital systems and expands e-prescribing to controlled substances (EPCS).

2. Problem Statement & Objectives

Patients in underserved and rural areas face long wait times and travel burdens to see specialists. Clinics lack a unified tool that combines scheduling, video visits, clinical documentation, and billing without juggling three or four separate vendors.

- Reduce average time-to-first-appointment from 12 days to under 48 hours.
- Achieve HIPAA-compliant, end-to-end encrypted video consultations at 99.9% session reliability.
- Enable clinicians to e-prescribe (including EPCS-certified controlled substances) without leaving the platform.
- Integrate with at least two major EHR systems (Epic, Cerner) via HL7 FHIR by year-end.

3. Target Users & Personas

Persona	Description	Primary Needs
Patient	Individual seeking virtual care	Easy booking, clear visit summaries, secure mes
Clinician	Physician, NP, or PA delivering care	Fast charting, e-prescribing, schedule manageme
Care Coordinator	Clinic staff supporting patients	Insurance verification, referral tracking
Compliance / Privacy Officer	Internal HIPAA compliance lead	Audit trails, access reviews, breach reporting

4. Scope

4.1 In Scope

- Patient self-scheduling with clinician availability sync.

- Real-time WebRTC video consultations with in-visit chat and screen share.
- Clinical documentation (SOAP notes) and structured e-prescribing (including EPCS).
- Insurance eligibility checks and claims submission.
- Secure asynchronous messaging between patient and care team.
- HL7 FHIR-based EHR integration for record exchange (Epic, Cerner).

4.2 Out of Scope (this release)

- In-home diagnostic device integration (e.g., remote vitals monitors).
- Group / family therapy session support.
- International (non-US) licensure and prescribing workflows.

5. Functional Requirements

5.1 Identity, Consent & Access

- FR-1: System shall verify patient identity at registration (KBA or document upload) before granting access to health records.
- FR-2: System shall capture and version informed-consent and telehealth-consent forms per state regulation.
- FR-3: System shall enforce minimum-necessary, role-based access to PHI for all internal users.

5.2 Scheduling & Visits

- FR-4: System shall allow patients to search clinicians by specialty, state licensure, and availability.
- FR-5: System shall support real-time video visits with automatic reconnection and a fallback audio-only mode.
- FR-6: System shall record visit metadata (duration, participants, connection quality) for billing and QA, but not video content by default.

5.3 Clinical Documentation & e-Prescribing

- FR-7: System shall provide structured SOAP note templates that map to ICD-10 and CPT codes.
- FR-8: System shall transmit prescriptions electronically to the patient's pharmacy of choice via a certified e-prescribing network.
- FR-9: System shall require two-factor identity proofing for clinicians before enabling EPCS (controlled substance) prescribing, per DEA requirements.

5.4 Billing & Insurance

- FR-10: System shall verify patient insurance eligibility in real time prior to visit confirmation.
- FR-11: System shall generate and submit claims (CMS-1500) to clearinghouses automatically post-visit.

5.5 Interoperability

- FR-12: System shall exchange patient records (CCD/C-CDA) with connected EHR systems via HL7 FHIR R4 APIs.
- FR-13: System shall reconcile medication lists between MediConnect and the external EHR to prevent conflicting prescriptions.

6. System Architecture Overview

MediConnect uses a layered microservices architecture (see attached Architecture Diagram). Patients and clinicians connect through web and mobile clients that terminate at a CDN/WAF and API gateway enforcing authentication and rate limiting. The Identity & Consent Service manages patient identity proofing, MFA, and consent capture, forming the entry point to a HIPAA-scoped segment that includes the Patient Records Service, Clinical Notes & e-Prescribing Service, and the Document/Imaging Store. The Video Consultation Service brokers WebRTC sessions via a third-party TURN/media relay rather than routing video through application servers. Billing and scheduling events flow through a shared event bus to keep the Insurance/Billing Service and Notification Service eventually consistent with visit state.

- Client Layer: Patient Mobile App, Patient Web Portal, Clinician Web Console.
- Edge/Perimeter: CDN/WAF, API Gateway (authn, rate limiting), Load Balancer.
- Application Services: Identity & Consent Service (SSO/MFA), Appointment Scheduling Service, Video Consultation Service (WebRTC), Clinical Notes & e-Prescribing Service, Billing & Insurance Service, Notification Service.
- Data Stores: Patient Records DB (PostgreSQL, PHI, encrypted at rest), Document/Imaging Store (S3, DICOM/PDF), Redis Session Cache, HIPAA Audit Log Store, Message/Event Bus.
- External Dependencies: EHR Systems (Epic/Cerner via HL7 FHIR), Payment Processor (Stripe), Pharmacy Network (Surescripts e-Prescribing), Insurance Eligibility API (Availity), Video/TURN Relay Provider.

7. Data Classification & Sensitive Data Inventory

Data Type	Classification	Where Stored	Regulatory Driver
Patient demographics & contact info	PHI / PII	Patient Records DB, encrypted at rest	HIPAA
Clinical notes & diagnoses	PHI	Patient Records DB + Document Store	HIPAA
Prescription data (incl. controlled substances)	PHI / Restricted	Clinical Notes Service, e-Rx network	HIPAA, DEA / EPCS
Insurance & payment details	PHI / Financial	Billing Service, tokenized via processor	HIPAA, PCI DSS
Video visit metadata	PHI (limited)	Video Consultation Service logs	HIPAA
Identity verification documents	Confidential / PII	Encrypted object storage	HIPAA, state privacy law

8. Authentication, Authorization & Session Management

- Patients authenticate via email/password or SSO with mandatory step-up MFA (SMS or authenticator app) before viewing records.
- Clinicians require MFA plus periodic re-verification of state medical license status; EPCS prescribing requires a separate identity-proofed credential per DEA rules.
- Role-based access control enforces minimum-necessary access: Clinicians see only their assigned patients; Care Coordinators see scheduling/billing but not clinical notes.
- All internal service-to-service calls use mutual TLS; every PHI read/write is logged to an immutable, query-able HIPAA audit trail.
- Session tokens expire after 15 minutes of inactivity for clinical interfaces; video sessions use short-lived, single-use room tokens.

9. Non-Functional Requirements

Category	Requirement
Availability	99.9% uptime for scheduling and video services; multi-AZ deployment
Video Quality	P95 video join time under 3 seconds; automatic bitrate adaptation
Scalability	Support 5,000 concurrent video visits during peak flu-season demand
Data Retention	Clinical records retained per state law (typically 7-10 years)
Auditability	All PHI access logged with actor, timestamp, and justification for 6+ years

10. Compliance & Regulatory Requirements

- HIPAA Security Rule and Privacy Rule compliance, including signed Business Associate Agreements with all subprocessors.
- 42 CFR Part 2 protections for behavioral health records where applicable.
- DEA EPCS certification for controlled-substance e-prescribing.
- State-specific telehealth practice and cross-state licensure requirements.
- SOC 2 Type II report required for enterprise health-system customers.

11. Assumptions, Constraints & Dependencies

- Clinician license verification is provided by a third-party credentialing vendor, not built in-house.
- EHR interoperability timeline depends on partner health systems' sandbox availability for FHIR certification testing.
- Video infrastructure relies on a third-party WebRTC/TURN relay provider for NAT traversal at scale.

12. Open Questions & Risks

- Should asynchronous (store-and-forward) visits be supported for dermatology use cases in a future release?
- Risk: EPCS identity-proofing adds friction to clinician onboarding; may slow provider network growth.
- Risk: FHIR data mapping inconsistencies across EHR vendors could cause medication list conflicts.